

EEE T202

CHINES & TRANSFORMERS

MODULE:3

1 – Speed Control of DC Motors

of operation of DC motors and select appropriate motor types for

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Syllabus

generation of torque, torque equation, performance characteristics – starters – 3 point and 4 point starters (principle only).

tors - field control, armature control. Braking of dc motors. Power and efficiency. Testing of dc motors - Swinburne's test, Hopkinson's test. DC motor applications – numerical problems.

Speed Control

$$E_b + I_a R_a$$

$$V - I_a R_a$$

$$= V - I_a R_a$$

$$E_b = \frac{Z\phi N}{60} \frac{P}{A} = K\phi\omega$$

$$\frac{V - I_a R_a}{K\phi}$$

rolling motor speed are
e (Voltage control)

armature circuit (Armature control)
rol)

Speed Control

is controlled to control the motor speed.

ed are in direct relation

ases, speed increases and vice versa

mature circuit is varied to control the speed

creases, speed decreases and vice versa

control the speed

are inversely proportional

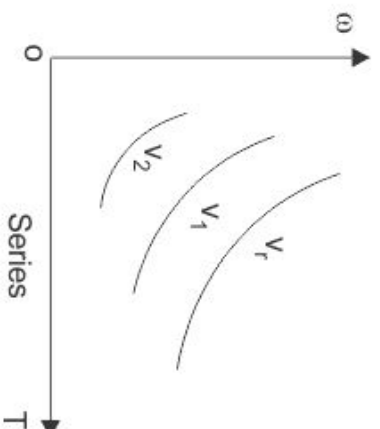
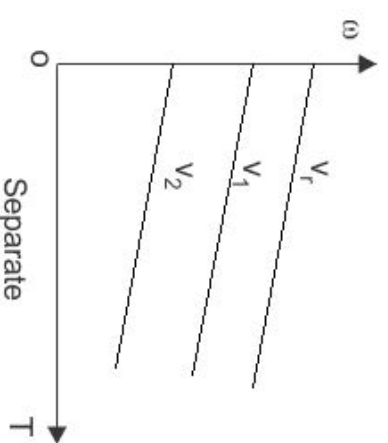
s, speed decreases and vice versa

$$\omega = \frac{V - I_a R_a}{K\phi}$$

Voltage Control

Speed is controlled by controlling the terminal voltage of the motor. This is directly proportional to the terminal voltage.

When the terminal voltage that can be applied across a dc motor will be maximum speed that can be attained by this method is the maximum speed that can be attained for the speed control of motor below this method is suitable for the speed control of motor below



Armature Voltage Control V_r (rated) $> V_1 > V_2$

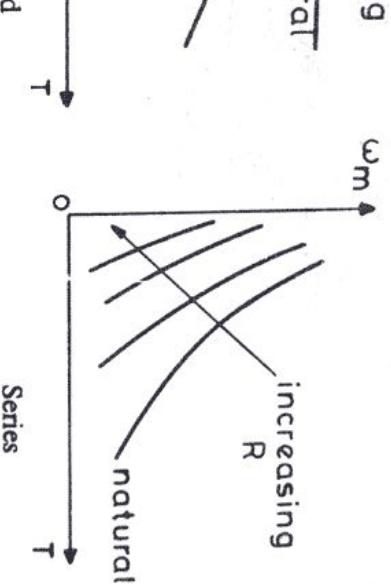
Armature Control

When speed control below base speed is required.

It is directly connected in series with the armature of dc

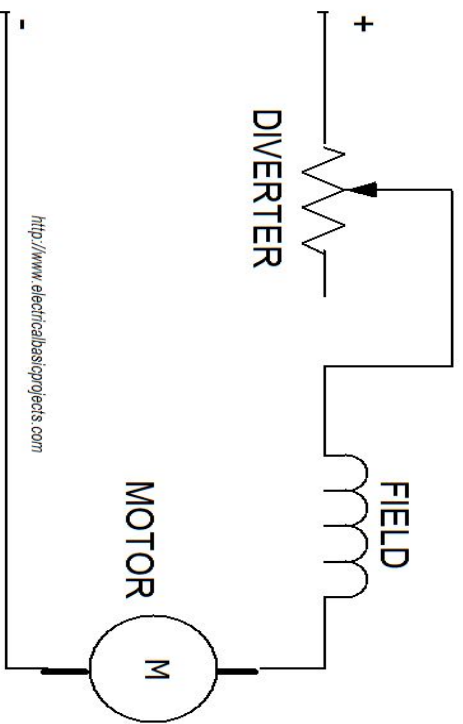
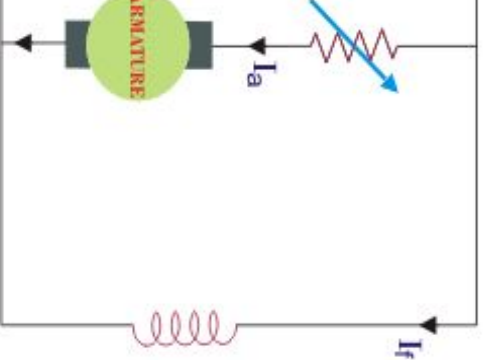
motor, wasting power in this external resistor.

It is an inefficient method of speed control, this method is not

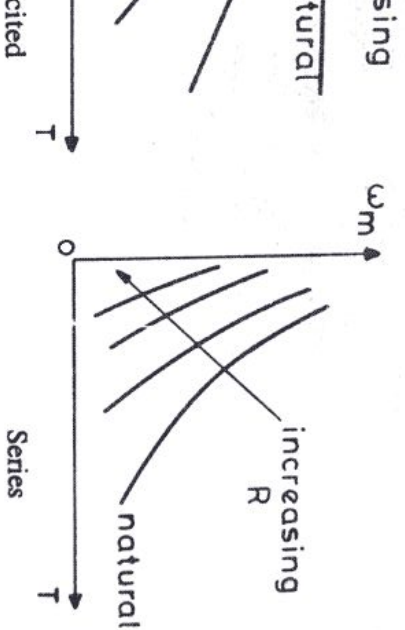


$$\omega = \frac{V - I_a R_a}{K\phi}$$

Armature Control



<http://www.electricalbasicsprojects.com>



$$\omega = \frac{V - I_a R_a}{K\phi}$$

Field Control

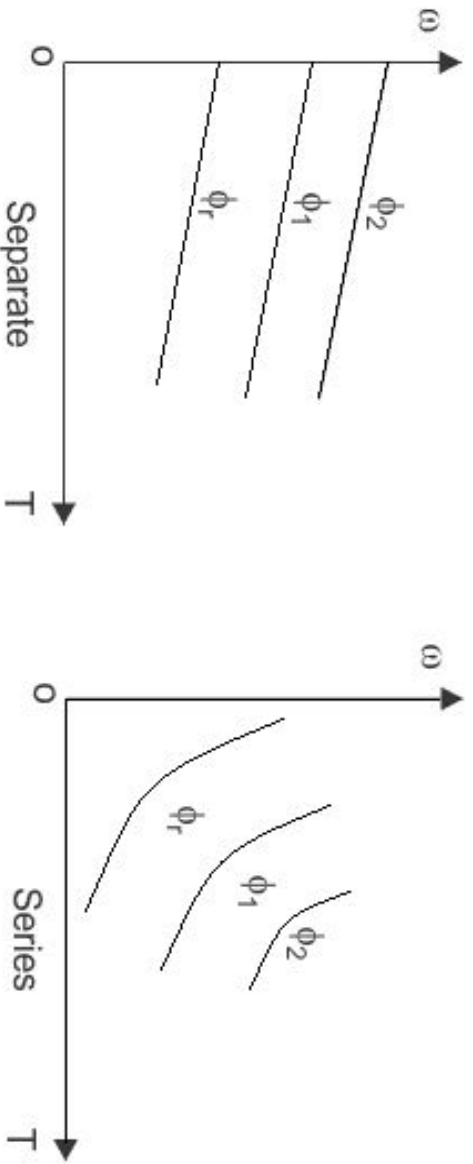
$$\omega = \frac{V - I_a R_a}{K\phi}$$

inversely proportional to the speed of motor, by varying control the speed of motor.

flux we can increase the motor speed and by increasing rease the speed.

field flux is not preferred. It is because of the saturation of

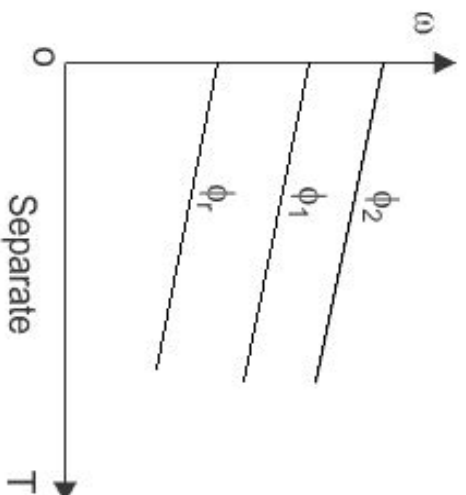
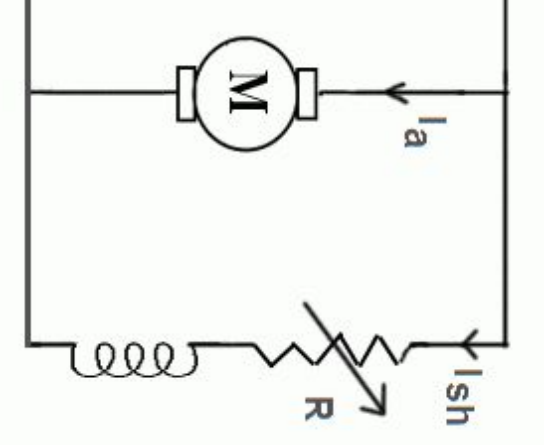
field flux control, speed of the motor is controlled only



Field flux control ϕ_r (rated) $> \phi_1 > \phi_2$

Field Control

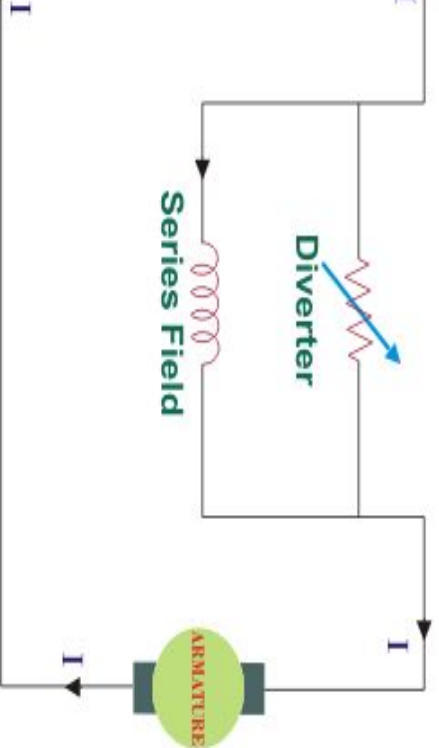
$$\omega = \frac{V - I_a R_a}{K\phi}$$



Field Control

$$\omega = \frac{V - I_a R_a}{K\phi}$$

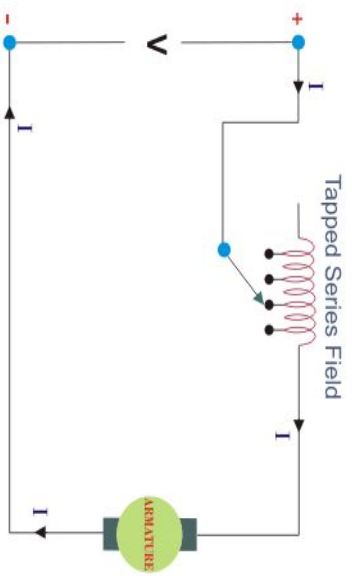
variable resistance (called field diverter) is connected in field winding. It takes some portion of the line current and winding, thus weakening the field and increasing the motor carries zero current, motor runs at minimum speed (called speed). When diverter resistance decreases, current decreases and speed of motor increases.



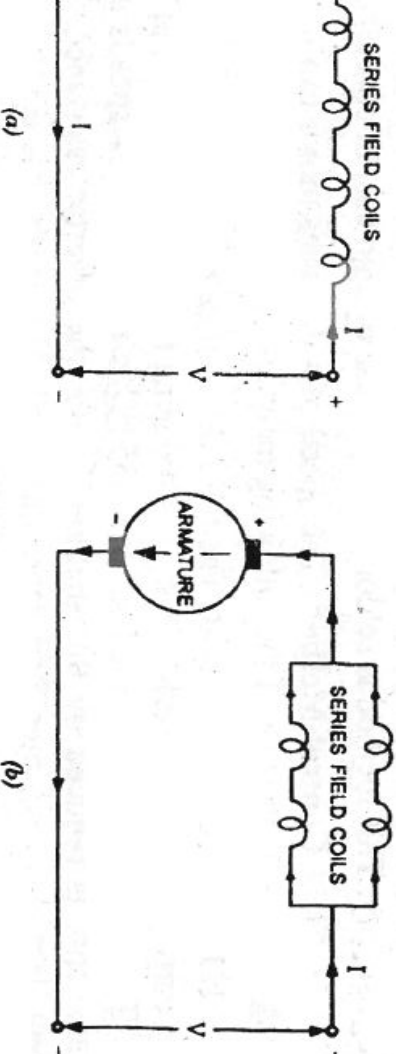
Field Control

Control: In this method flux is controlled by changing the number of turns in the field winding. By means of providing tapings. By changing the number of turns, flux also can be increased or decreased. With full winding, the motor runs at base speed.

$$\omega = \frac{V - I_a R_a}{K\phi}$$



connection of field coils: In this method, field coils are connected in series and parallel connections of coils. Regrouping will give one effective value of inductance and thus the field flux can be attained.



Problems

A motor has armature resistance of $0.25\ \Omega$, on load it takes an armature current of $50\ \text{A}$ and runs at $750\ \text{rpm}$. If the flux of the motor is constant without changing the load torque, find the new speed of the motor.

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Problems

or runs at a normal speed of 250 rpm when the armature resistance of armature is 0.12 ohm. Calculate the speed resistance is connected in the field winding to reduce the normal flux and armature current is 100 A.

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Problems

motor running at 1000 rpm takes an armature current of 100 A. If the motor is to be run at 600 rpm, what must be the value of the armature current to reduce the speed to 600 rpm. What must be the value of the field current if the original armature current is 100 A and the field current is 5 A. Take armature current is constant during this process.

motor running at 1000 rpm takes an armature current of 100 A to reduce the speed to 600 rpm. What must be the value of the resistance inserted in the armature circuit if the original armature current is 100 A? Assume the field current is constant. Find the armature current when the speed is 600 rpm. Take armature current is constant during this process.

Problems

Motor has armature and field resistances of 1.2 ohm and 500 ohm. When running on no-load, the current taken is 4 A and the speed is 1200 rpm. Calculate the speed when motor is fully loaded and the armature current is 26 A. The supply is 250 V.

motor has armature and field resistances of 1.2 ohm and 500 ohm respectively. Calculate the speed when motor is fully loaded and the current taken from the supply is 26 A.